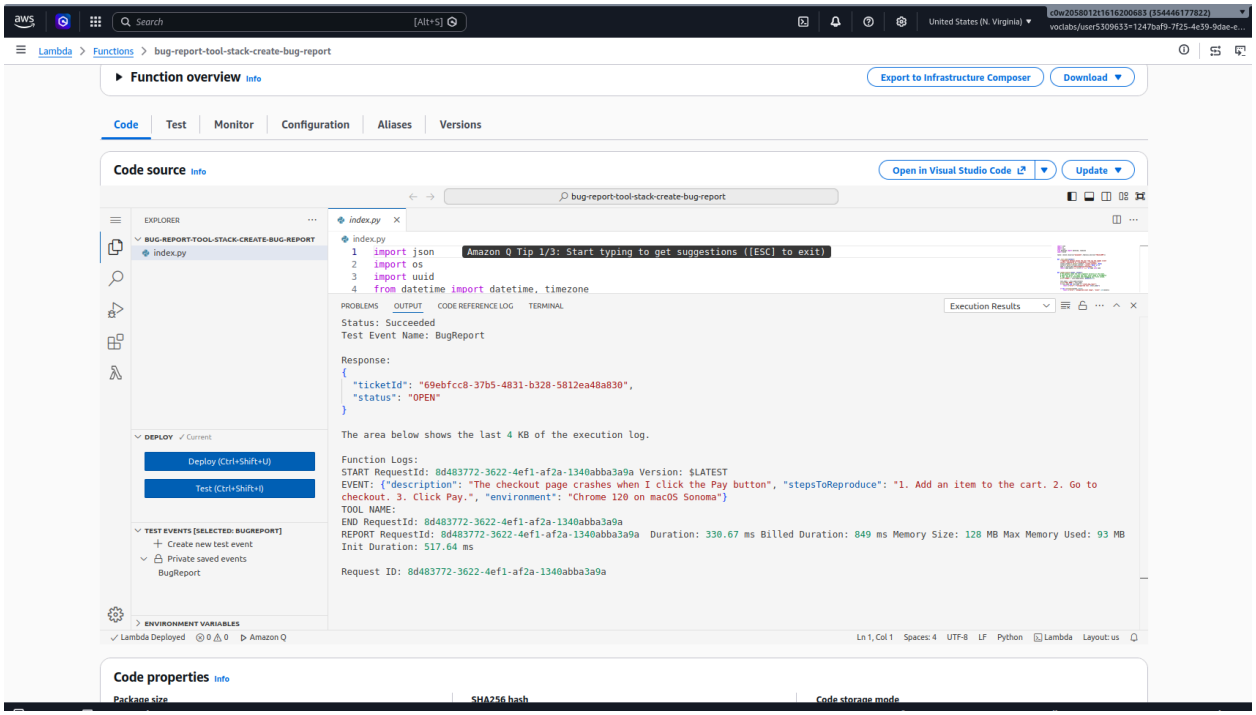
Customer Support Chatbot with Amazon Bedrock AgentCore

Lambda Function Isolation Test

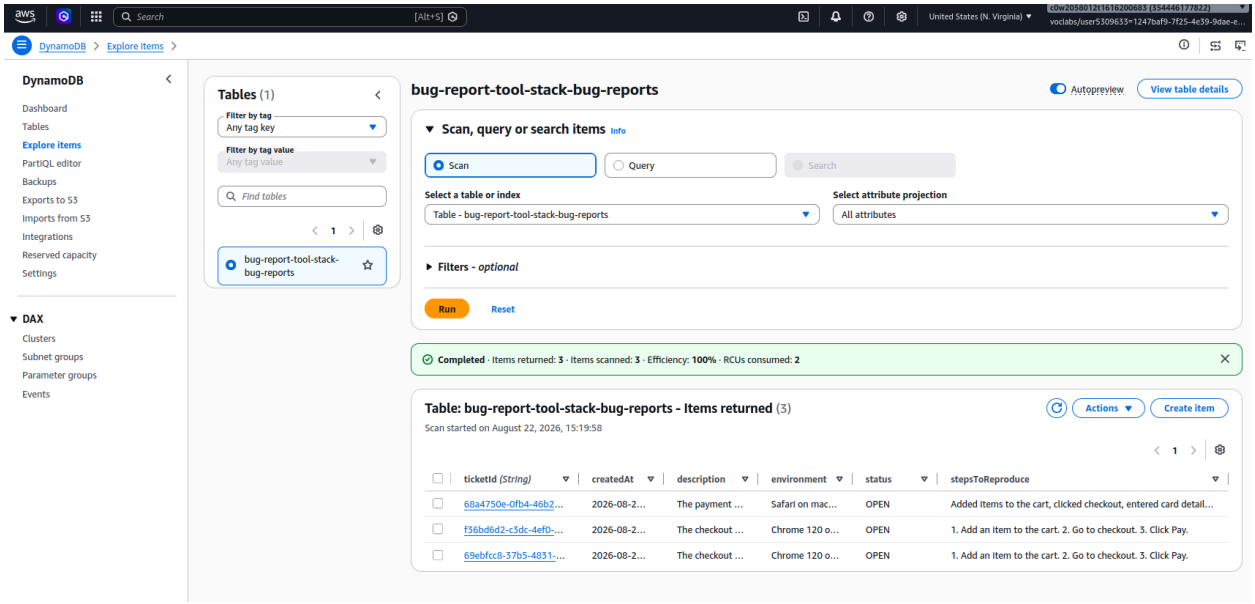
The Lambda function **bug-report-tool-stack-create-bug-report** handles creating bug tickets in DynamoDB. To ensure it operates correctly in isolation, a test JSON event was sent directly via the AWS Lambda Console.



Screenshot 1: Lambda test execution result showing a green **Succeeded** status and the generated ticketId

DynamoDB Data Persistence Check

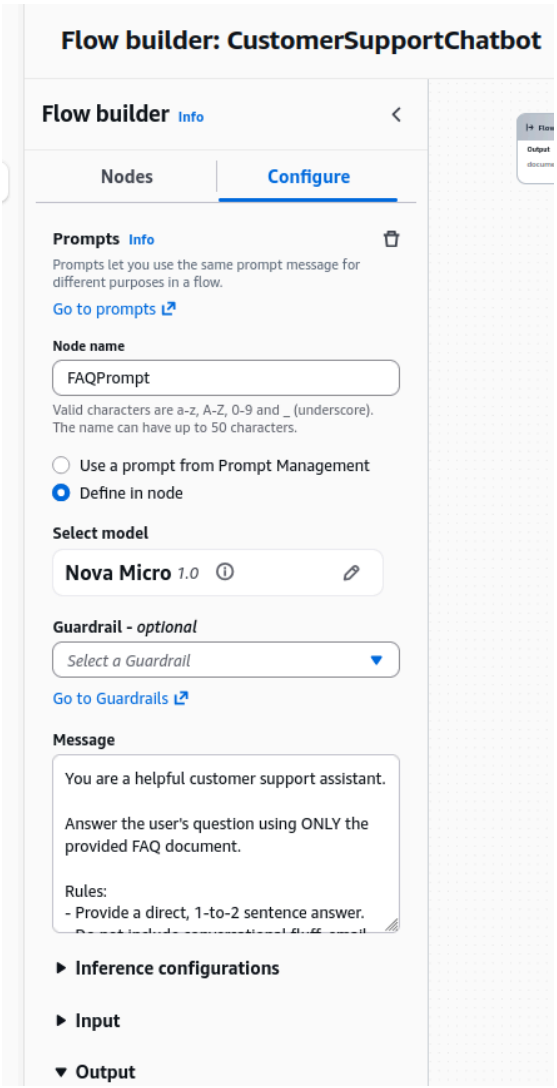
Following the Lambda test, we inspected the DynamoDB table bug-report-tool-stack-bug-reports to confirm that the payload was written correctly.



Screenshot 2: DynamoDB table item view displaying the created ticket

FAQ Prompt Node Template Configuration

The `FAQPrompt` node handles customer questions using the embedded FAQ knowledge base. It is configured with specific guardrails and instructions to either answer directly or redirect to human support if the request is not covered.



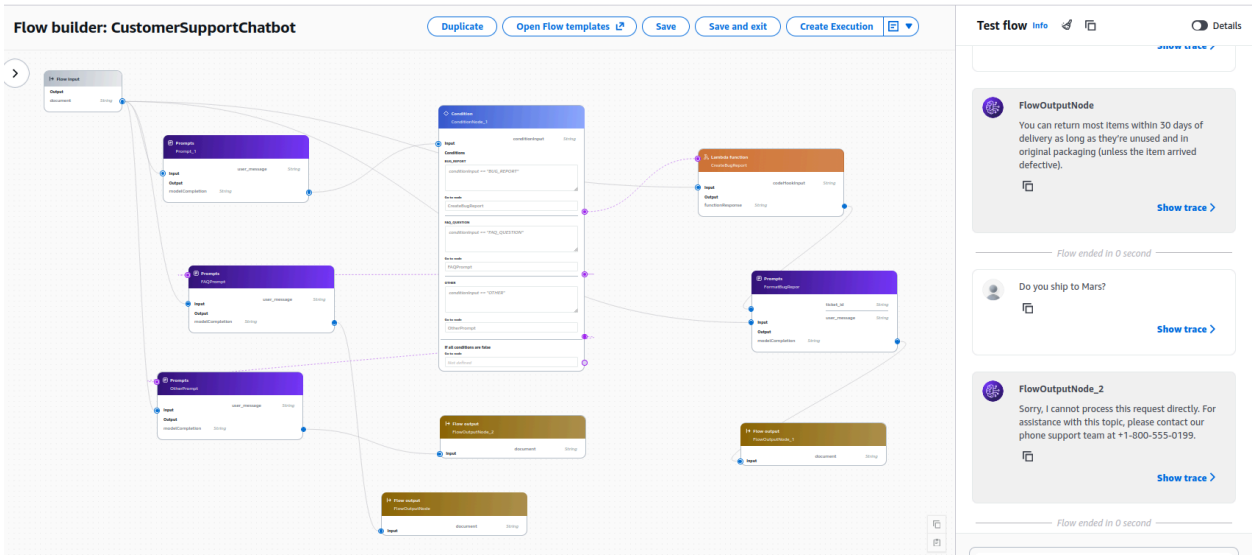
Screenshot 3: AWS Bedrock Flow `FAQPrompt` node configuration panel showing the system prompt template, model selection (`Nova Micro 1.0`), and rules.

Flow Test Response: Covered FAQ Question

When a user asks a question that exists within the FAQ database (e.g., return policies), the flow correctly matches the prompt context and returns a direct answer.

Flow Test Response: Uncovered FAQ Question

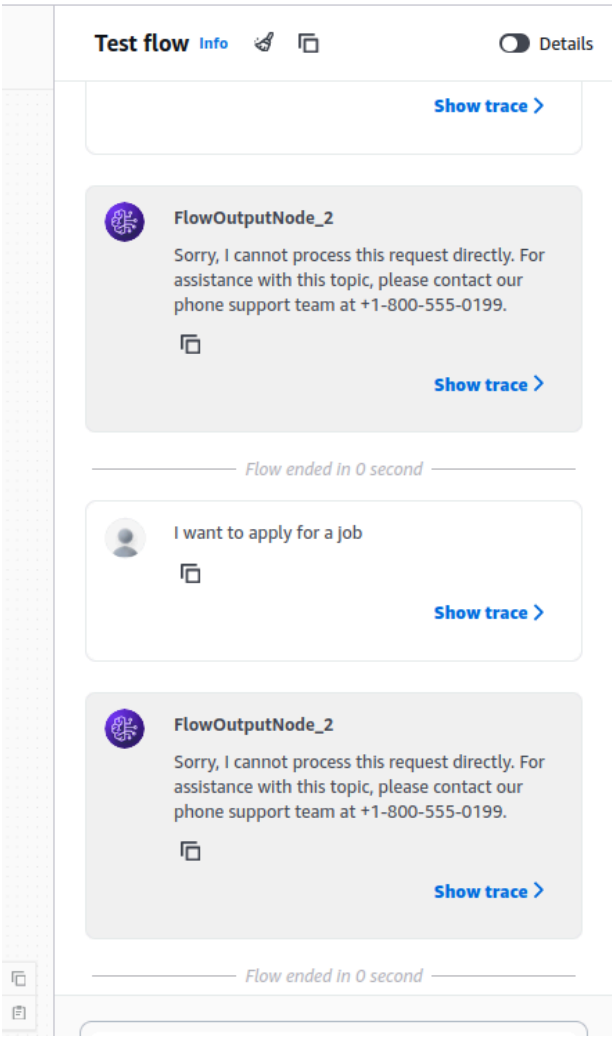
When a user asks a platform question that is NOT covered by the provided FAQ (e.g., *"Do you ship to Mars?"*), the model follows system instructions and directs the user to human phone support



Screenshot 4: Bedrock Flow execution test for a covered question (*"You can return most items within 30 days..."*) and Bedrock Flow execution test for an uncovered FAQ question returning the support phone number (`+1-800-555-0199`).

Flow Test Response: Other Customer Support Requests

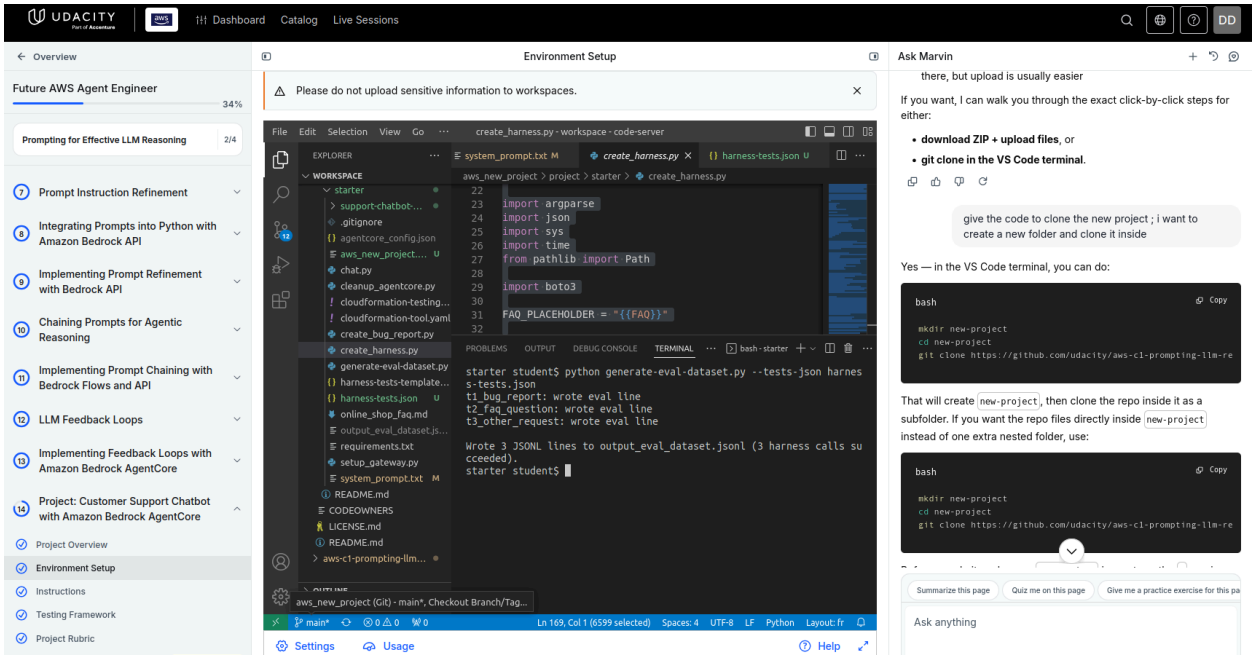
For requests categorized under the `OTHER` path (e.g., job applications, partnerships), the flow routes the query through the `OtherPrompt` node, which directs the user to human support.



Screenshot 5: Bedrock Flow execution test for an out-of-scope request (*"I want to apply for a job"*) redirecting to the support phone number (`+1-800-555-0199`).

Live Chat Session Test

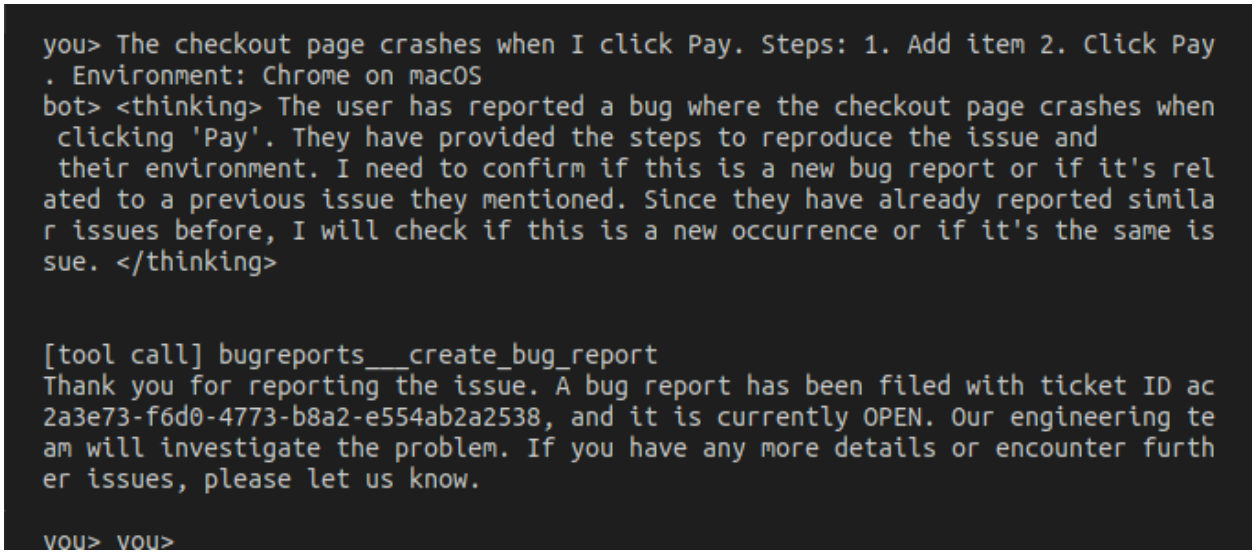
After deploying the agent harness configured with our system prompt, an interactive terminal chat session was conducted to test real-time tool calling and conversational flow



Screenshot 6 : Interactive CLI chat session terminal execution (python chat.py)

Evaluation Dataset Generation

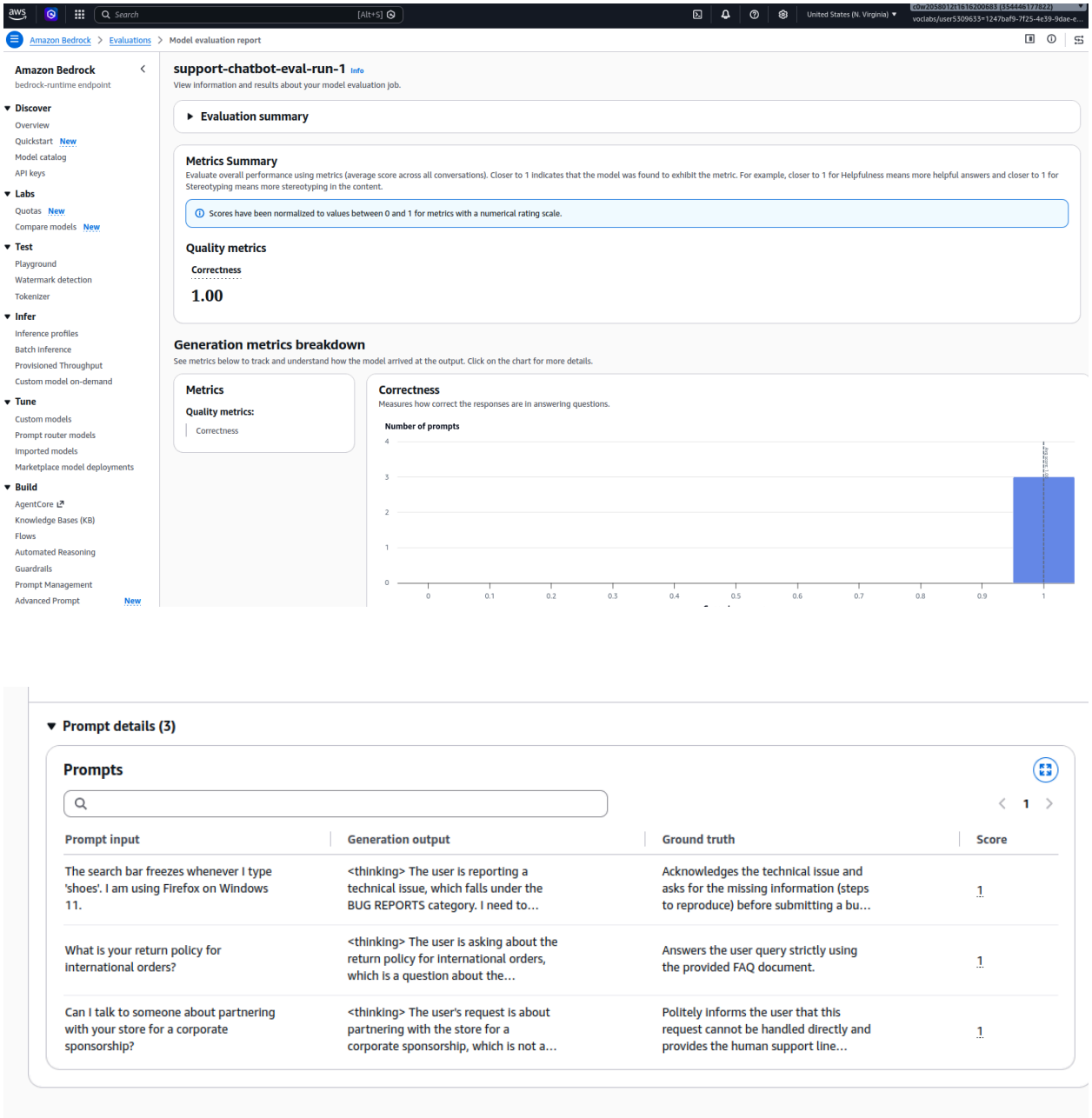
Using our test suite (harness-tests.json), we generated the JSONL dataset covering bug reporting, FAQ lookup, and out-of-scope scenarios.



Screenshot 7: Terminal execution output of python generate-eval-dataset.py --tests-json harness-tests.json displaying (3 harness calls succeeded).

LLM-as-a-Judge Evaluation

The generated evaluation dataset was passed to the LLM-as-a-Judge execution step to score model alignment against expected outcomes.



Screenshot 8: LLM-as-a-Judge evaluation results showing passing scores across test cases.